

***SIMPLIFIED AIRCRAFT
TURBULENCE RISK ESTIMATOR***
(PYTHON-BASED PROJECT)

**Submitted by:
Prarthana R S
B.Tech Aerospace Engineering
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INTRODUCTION

Aircraft turbulence is a familiar atmospheric occurrence and one that can compromise the stability of an aircraft in flight. Such conditions are subject to a number of variables, from wind speed and shear to changes in temperature, as well as the altitude and speed of the aircraft itself.

To get at the root of what drives these factors, we have put together the Simplified Aircraft Turbulence Risk Estimator, a Python project for their analysis. The program takes in the relevant flight and atmospheric data and runs some elementary calculations to produce a set of figures: air density, pressure, sound speed, Mach number, flight regime, wind shear and so on. From there it will assign a turbulence score and put it in one of three categories – LOW, MODERATE or HIGH.

In essence, the project demonstrates how one can apply straightforward Python to an aerospace engineering challenge. It should be noted, however, that this is strictly an educational tool and is not meant to provide real-time weather forecasts for aviation.

PROBLEM STATEMENT

Various atmospheric and flight conditions affect the turbulence of the aircraft such as **windspeed, windshear, temperature variation, aircraft speed**, etc.

The main issue being solved by this project is the development of a simple Python program which will generate a turbulence value based on various atmospheric and flight conditions.

OBJECTIVES

- For creating a turbulence risk calculator based on Python.
- For accepting aircraft and atmospheric inputs from the user.
- For calculating the atmospheric variables.
- For calculating the Mach number and defining the flight regime.
- For calculating the wind shear.
- For integrating some selected variables to form a turbulence score.
- For categorizing the risk into LOW, MODERATE, and HIGH.

FUNCTIONAL REQUIREMENTS

Atmospheric Calculations

The system performs the following calculation: The functional requirements list the primary functions of the Simplified Aircraft Turbulence Risk Estimator.

User Input

The program accepts the following values as inputs from the user:

- Initial aircraft altitude
- Final aircraft altitude
- Aircraft speed
- Temperature change
- Initial wind speed
- Final wind speed
- Wind speed change

Input Validation

Basic validation is done to detect any wrong input such as negative altitude, negative speed of the aircraft, and negative speed of the wind.

- Atmospheric temperature
- Atmospheric pressure
- Air density
- Speed of sound

Mach Number and Flight Regime

The Mach number is determined using the speed of the aircraft and speed of sound. Then the flight regime is classified as subsonic, transonic, supersonic, or hypersonic.

Wind Shear Calculation

Wind shear is calculated based on the wind speed variation and altitude variation.

Turbulence Score

The turbulence rating is determined by measuring certain parameters against threshold values.

Risk Classification

Given the turbulence score, the program will categorize the estimated risk level as:

- LOW
- MODERATE
- HIGH

Display of Results

These values are displayed on the screen using the console interface. This can be considered an implementation of the high-level features mentioned above for the program.

NON-FUNCTIONAL **REQUIREMENTS**

These non-functional requirements detail how the project will work as well as its characteristics.

Usability

The interface for this program needs to be straightforward and user-friendly. The user will simply input the necessary data for the plane and atmosphere and view the output.

Reliability

The algorithm must generate consistent outputs upon entering identical inputs. Basic validation of inputs is incorporated to minimize mistakes due to erroneous inputs.

Performance

The calculations should be done fast because the program only does math and basic atmospheric work. It doesn't handle tasks so results come out quickly. This makes it easy to use for checks or basic analysis. No need to wait long for answers.

Maintainability

The Python code should stay simple and easy to understand. This way students can read it change it and make it better on.

Error Handling

The program terminates and shows an error message in the event of particular invalid situations.

Educational Use

The design emphasizes simplicity, which allows the viewer to be exposed to a fundamental understanding of programming and aerospace concepts.

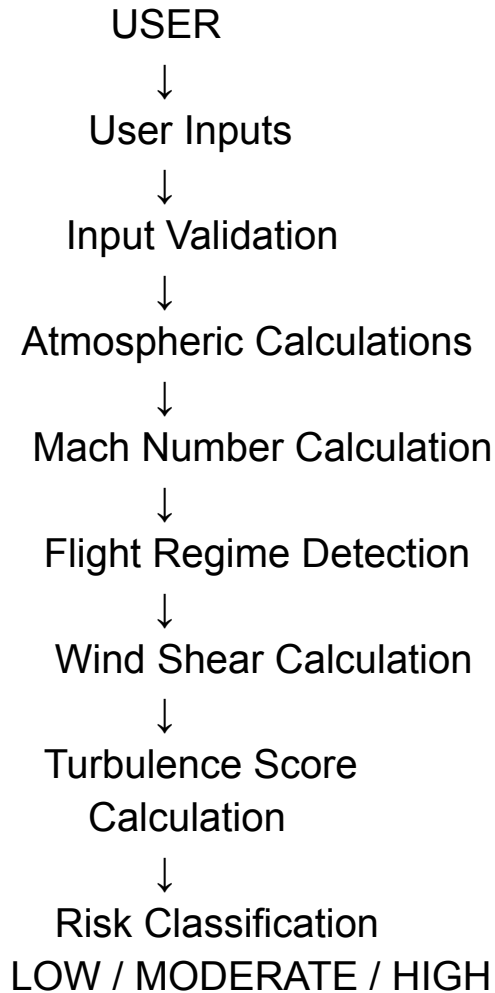
Resource Efficiency

The program is run locally and does not require a database or huge amount of external data.

SYSTEM ARCHITECTURE

The Simplified Aircraft Turbulence Risk Estimator has a simple sequential structure. The user enters the aircraft and atmospheric data, the program checks the entered information, performs calculations, determines the turbulence score, and displays the estimated risk level.

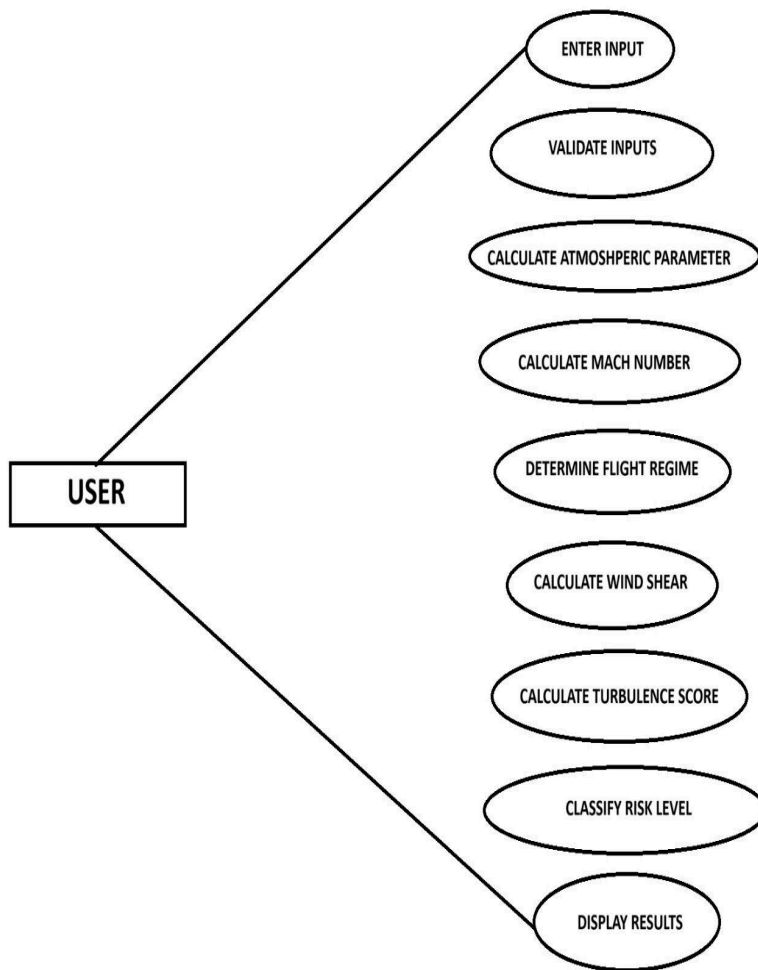
ARCHITECTURE FLOW



DESIGN DIAGRAM

CASE DIAGRAM

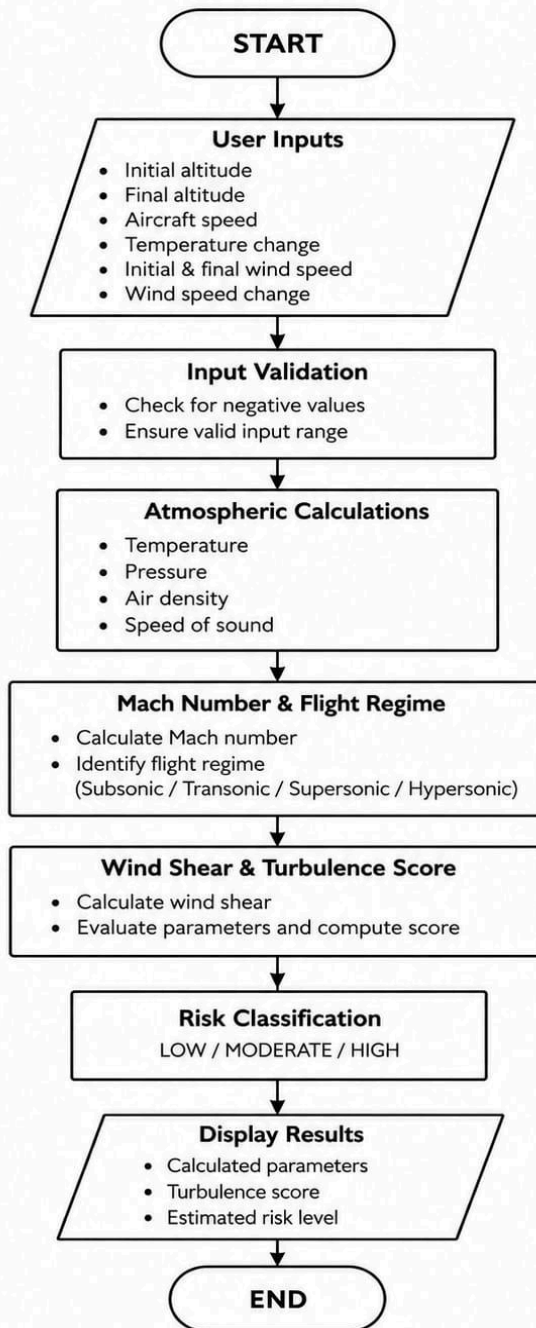
The use case diagram below represents the interaction between the user and the simplified aircraft turbulence risk estimator. It demonstrates the basic operations of the considered software, ranging from inputting the required data to obtaining the estimated risk level.



WORKFLOW DIAGRAM

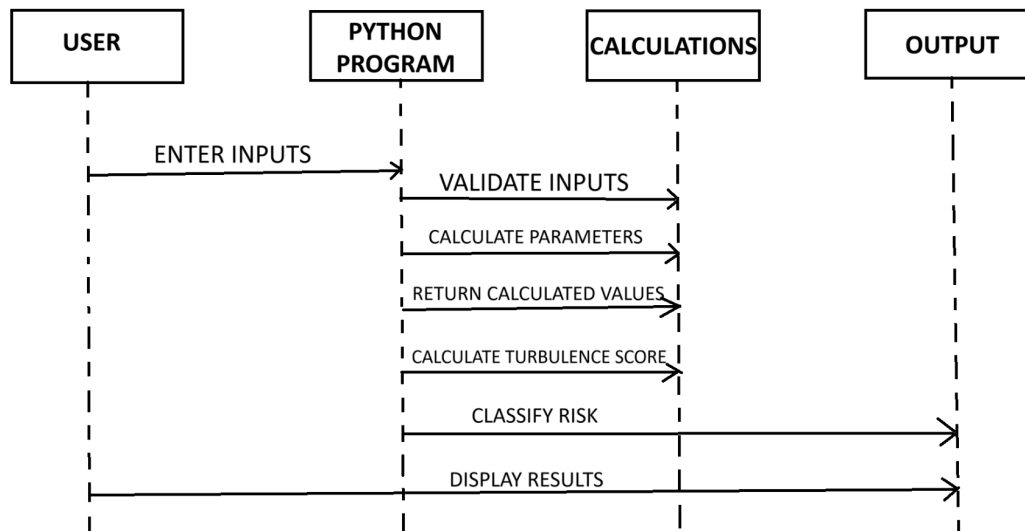
The workflow diagram below represents the program's steps, starting from inputting the data to the final risk assessment stage.

System Architecture of the Simplified Aircraft Turbulence Risk Estimator



SEQUENCE DIAGRAM

The sequence diagram below represents the interaction between the user and the program's logical stages. The current prototype is a simple Python script, and thus the stages are not organized into separate Python classes.



CLASS/COMPONENT DIAGRAM

The system is designed using a modular component-based structure, where each Python file performs a specific task. The main.py module is the central controller, while other modules are coordinated input_handler.py collects the required aircraft and atmospheric inputs, validation.py checks if the entered values are valid, and constants.py contains fixed atmospheric constants that are used by the atmosphere.py module to calculate the temperature, pressure, air density, and speed of sound.

The `flight_analysis.py` module calculates the Mach number, flight regime, and wind shear, and the `scoring.py` module calculates the turbulence score. The `risk.py` module converts the score into LOW, MODERATE, or HIGH risk, while `output.py` displays the calculated results to the user. This approach makes the program easier to comprehend and maintain, test, analyze, and modify.

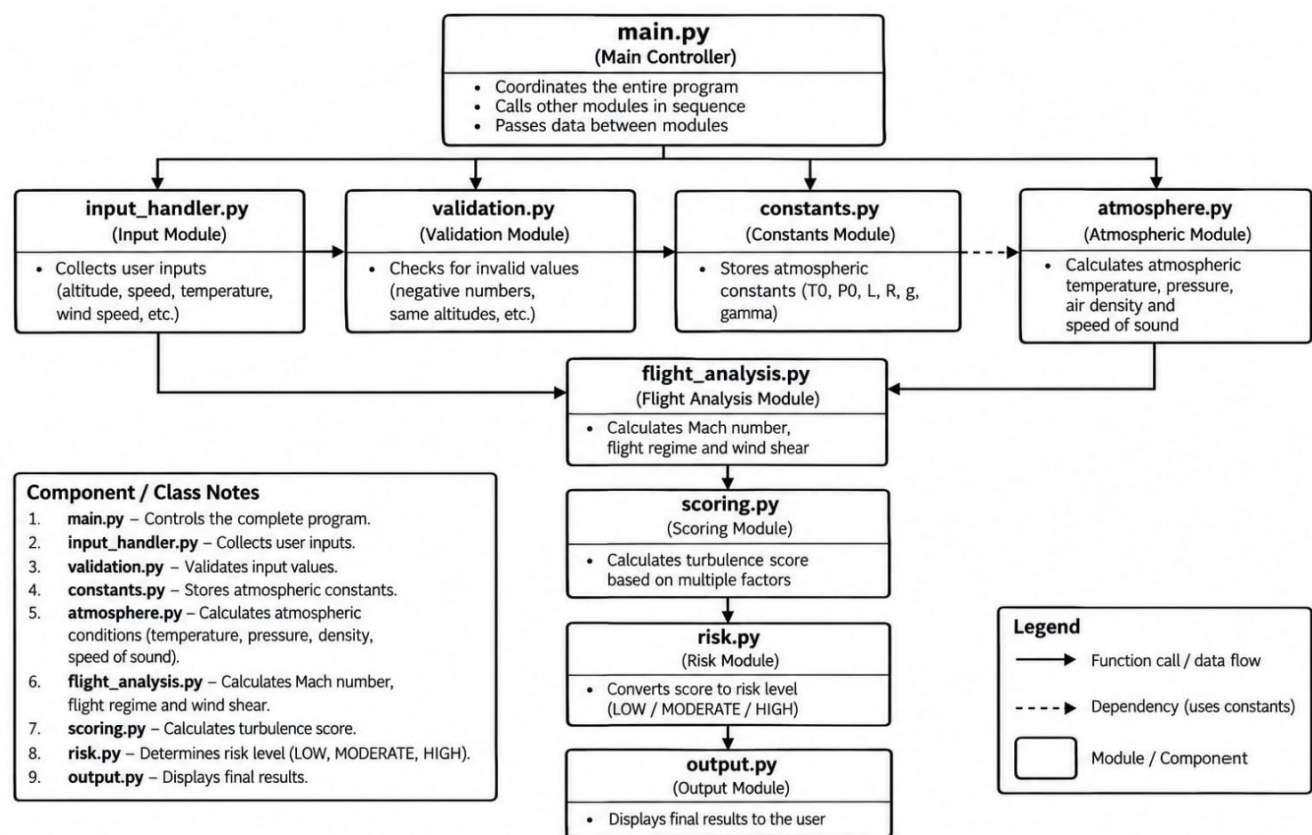


Figure 7.4: Class/Component Diagram of the Simplified Aircraft Turbulence Risk Estimator

DESIGN DECISIONS AND RATIONALE

DECISION	REASON
Command-Line Interface	A command line interface was chosen because the primary goal is to showcase calculations and Python logic, not add any extra GUI layer of complexity
Python	Python was selected due to its simple syntax and built-in math module, which will be used for calculation-heavy code
Parameter-Based Estimation	The program utilizes user input parameters instead of live weather data because the project is intended to be self-contained and simple enough for an introductory-level project
Score-Based Risk Classification	A score is a convenient way to tie multiple factors into a single, easily understandable value: LOW, MODERATE or HIGH
Basic Input Validation	Validation was added to ensure simple invalid inputs are caught before calculation
Simplified Model	The model is intentionally simple because the project's goal is educational and not an operational aviation risk assessment tool

IMPLEMENTATION DETAILS

The project was made using Python 3 and a math module. The program is being executed locally on either PowerShell or Command prompt.

INPUTS

The program takes in initial altitude, final altitude, aircraft speed, temperature change, initial wind speed, final wind speed and wind speed change as inputs.

ATMOSPHERIC CALCULATIONS

Atmospheric temperature, pressure, air density and speed of sound are computed using the various constants and formulas contained within the program.

MACH NUMBER AND FLIGHT REGIME

Mach number is determined by dividing an aircraft's speed by the speed of sound that was calculated. Then, based on the obtained value, it is possible to determine whether an aircraft is in a subsonic, transonic, supersonic, or hypersonic regime.

WIND SHEAR

Wind shear is determined by dividing the difference in wind speed by the difference in height; thus, wind shear is the absolute value of this ratio.

TURBULENCE SCORE AND RISK

The program analyzes such parameters as wind shear, final wind speed, wind speed change, temperature change, altitude, and Mach number. All of these variables have assigned thresholds; thus, when any of these thresholds are met, a point is given to the corresponding variable. Finally, based on the sum of points, the risk level is determined as LOW, MODERATE, or HIGH.

SCREENSHOTS AND RESULTS

The program was executed three times using different combinations of input. The three tests were performed to demonstrate LOW, MODERATE and HIGH expected risk outputs.

LOW RISK TEST

PARAMETER	VALUE
Initial altitude	2000 m
Final altitude	2500 m
Aircraft speed	150 m/s
Temperature change	2°C
Initial wind speed	5 m/s
Final wind speed	7 m/s
Wind speed change	2 m/s
Mach Number	0.451
Flight Regime	Subsonic
Wind Shear	0.004 s ⁻¹
Turbulence Score	5
Estimated Risk	LOW

```
Windows PowerShell
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PS C:\Users\prart\OneDrive\Documents\Simplified Aircraft Risk Estimator> python main.py
-----SIMPLIFIED AIRCRAFT TURBULENCE RISK ESTIMATOR-----
Enter aircraft altitude(1) (m):2000
Enter aircraft altitude(2) (m):2500
Enter aircraft speed (m/s):150
Enter temperature change(C):2
Enter initial wind speed(1)(m/s):5
Enter final wind speed(2)(m/s):7
Enter wind speed change(m/s):2
-----RESULT-----
Altitude           : 2500.0 m
Atmospheric Temperature: 275.15 K
Atmospheric Pressure   : 79488.42 Pa
Air Density          : 1.01 Kg/m^3
Speed of Sound         : 332.53 m/s
Aircraft Speed        : 150.0 m/s
Mach Number           : 0.451
Flight Regime          : Subsonic
Wind Speed             : 7.0 m/s
Wind Speed Change      : 2.0 m/s
wind Shear             : 0.004 s^-1
Turbulence Score       : 5
Temperature Change     : 2.0 C
Estimated Risk         : LOW
PS C:\Users\prart\OneDrive\Documents\Simplified Aircraft Risk Estimator> |
```

MODERATE RISK TEST

PARAMETER	VALUE
Initial altitude	5000 m
Final altitude	6000 m
Aircraft speed	180 m/s
Temperature change	10°C
Initial wind speed	10 m/s
Final wind speed	20 m/s
Wind speed change	3 m/s
Mach Number	0.562
Flight Regime	Subsonic
Wind Shear	0.01 s ⁻¹
Turbulence Score	10
Estimated Risk	MODERATE

```
Windows PowerShell
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PS C:\Users\prart\OneDrive\Documents\Simplified Aircraft Risk Estimator> python main.py
-----SIMPLIFIED AIRCRAFT TURBULENCE RISK ESTIMATOR-----
Enter aircraft altitude(1) (m):5000
Enter aircraft altitude(2) (m):6000
Enter aircraft speed (m/s):180
Enter temperature change(C):10
Enter initial wind speed(1)(m/s):10
Enter final wind speed(2)(m/s):20
Enter wind speed change(m/s):3
-----RESULT-----
Altitude           : 6000.0 m
Atmospheric Temperature: 255.65 K
Atmospheric Pressure  : 54007.94 Pa
Air Density         : 0.74 Kg/m^3
Speed of Sound       : 320.53 m/s
Aircraft Speed       : 180.0 m/s
Mach Number          : 0.562
Flight Regime         : Subsonic
Wind Speed            : 20.0 m/s
Wind Speed Change     : 3.0 m/s
wind Shear            : 0.01 s^-1
Turbulence Score      : 10
Temperature Change    : 10.0 C
Estimated Risk        : MODERATE
PS C:\Users\prart\OneDrive\Documents\Simplified Aircraft Risk Estimator> |
```

HIGH RISK TEST

PARAMETER	VALUE
Initial altitude	5000 m
Final altitude	7000 m
Aircraft speed	200 m/s
Temperature change	15°C
Initial wind speed	10 m/s
Final wind speed	20 m/s
Wind speed change	5 m/s
Mach Number	0.624
Flight Regime	Subsonic
Wind Shear	0.005 s ⁻¹
Turbulence Score	11
Estimated Risk	HIGH

```
Windows PowerShell
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PS C:\Users\prart\OneDrive\Documents\Simplified Aircraft Risk Estimator> python main.py
-----SIMPLIFIED AIRCRAFT TURBULENCE RISK ESTIMATOR-----
Enter aircraft altitude(1) (m):5000
Enter aircraft altitude(2) (m):7000
Enter aircraft speed (m/s):200
Enter temperature change(c):15
Enter initial wind speed(1)(m/s):10
Enter final wind speed(2)(m/s):20
Enter wind speed change(m/s):5
-----RESULT-----
Altitude           : 7000.0 m
Atmospheric Temperature: 255.65 K
Atmospheric Pressure  : 54007.94 Pa
Air Density         : 0.74 Kg/m^3
Speed of Sound       : 320.53 m/s
Aircraft Speed       : 200.0 m/s
Mach Number          : 0.624
Flight Regime         : Subsonic
Wind Speed           : 20.0 m/s
Wind Speed Change     : 5.0 m/s
wind Shear            : 0.005 s^-1
Turbulence Score      : 11
Temperature Change    : 15.0 C
Estimated Risk        : HIGH
PS C:\Users\prart\OneDrive\Documents\Simplified Aircraft Risk Estimator> |
```

TESTING APPROACH

Testing was performed by executing the program with various input data combinations. The program's correctness was validated by ensuring that it correctly accepted the input data, performed the calculation, produced the expected numeric score, and displayed the appropriate risk category.

TEST ID	PURPOSE	SCORE	EXPECTED RISK	OBTAINED RISK
T01	Low-condition test	5	LOW	LOW
T02	Moderate-condition test	10	MODERATE	MODERATE
T03	High-condition test	11	HIGH	HIGH

VALIDATION TEST

TEST ID	CONDITIONS	EXPECTED BEHAVIOUR
VT01	Negative altitude	Display an error and stop
VT02	Same initial and final altitude	Display an error and stop
VT03	Negative aircraft speed	Display an error and stop
VT04	Negative wind speed	Display an error and stop

CHALLENGES FACED

UNDERSTANDING THE ATMOSPHERIC CALCULATIONS

Initially, relating the given altitude to temperature, pressure, density, and speed of sound was challenging. However, going through the calculations assisted in establishing a clear relationship between altitude and the four variables.

MACH NUMBER

Since the Mach number depends on the speed of sound, it became necessary to calculate the atmospheric conditions first before proceeding to determine the Mach number.

WIND SHEAR

The wind shear computation involved a clear understanding of the relationship between wind speed and altitude.

RISK SCORING

There was quite a challenge in combining various parameters to come up with a score. A threshold approach was adopted to come up with an acceptable solution, which would make the interpretation of values easy.

INPUT VALIDATION

The program required sanity checks for basic invalid values so that incorrect inputs would not get passed into the calculations.

TESTING

Various combinations of values had to be tried to produce and verify LOW, MODERATE and HIGH results.

LEARNING AND KEY TAKEAWAYS

This project taught me about the connections between a programming problem and an aerospace engineering application, and it also helped me get more comfortable writing a Python program from input to output, instead of with individual examples.

- Using input() and converting user input into numerical values.
- Working with variables, constants and arithmetic operations.
- Using if, elif and else statements to make decisions.
- Using the math module for calculations.
- Calculating atmospheric and flight-related parameters.
- Understanding the basic idea of Mach number and flight regimes.
- Calculating wind shear from changes in wind and altitude.
- Creating a simple scoring and classification system.
- Adding basic input validation.
- Running and testing a Python program using PowerShell.
- Organizing a project for GitHub and preparing technical documentation.

FUTURE ENHANCEMENTS

- Connect this program to reliable real-time weather or atmospheric data.
- Provide a graphical user interface.
- Expand the list of parameters that describe the atmosphere, such as humidity and pressure gradients.
- Refining the model of turbulence with more sophisticated aerospace relations.
- Provide graphs that depict altitude versus pressure, temperature, and wind, as well as relationships between different types of turbulence.
- Develop automated test cases for a greater range of inputs.
- Segregate future versions of the program into specific functions or modules that can be expanded upon in the future.

REFERENCE

- 1)VITyarthi – Build Your Own Project: General Project Instructions & Submission Guidelines, VIT Bhopal University.
- 2)Python 3 Documentation – Python programming language and math module reference.

3)Project Source Code – main.py, Simplified Aircraft Turbulence Risk Estimator.

4)Project Statement – statement.md.

5)Project README – README.md.